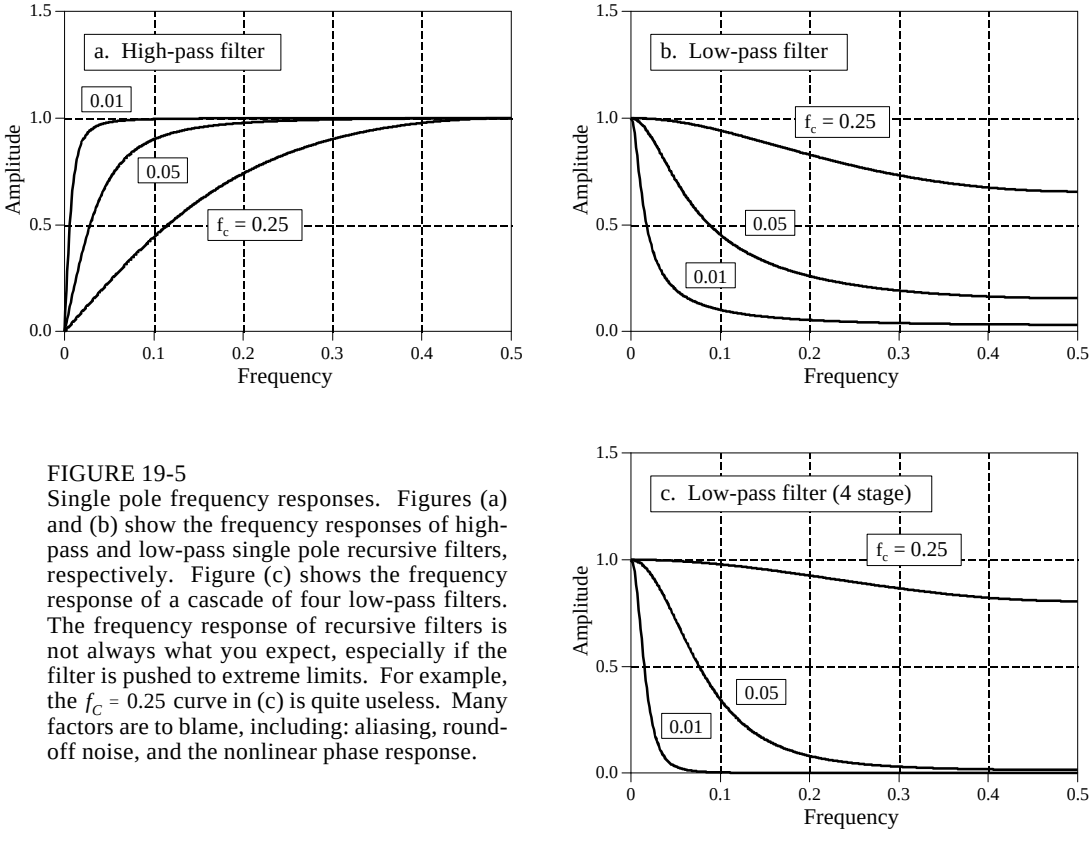

FIGURE 19-5
Single pole frequency responses. Figures (a) and (b) show the frequency responses of high-pass and low-pass single pole recursive filters, respectively. Figure (c) shows the frequency response of a cascade of four low-pass filters. The frequency response of recursive filters is not always what you expect, especially if the filter is pushed to extreme limits. For example, the f_C = 0.25 curve in (c) is quite useless. Many factors are to blame, including: aliasing, round-off noise, and the nonlinear phase response.

Figure 19-5 shows the frequency responses of various single pole recursive filters. These curves are obtained by passing a delta function through the filter to find the filter's impulse response. The FFT is then used to convert the impulse response into the frequency response. In principle, the impulse response is infinitely long; however, it decays below the single precision round-off noise after about 15 to 20 time constants. For example, when the time constant of the filter is d = 6.63 samples, the impulse response can be contained in about 128 samples.

The key feature in Fig. 19-5 is that single pole recursive filters have little ability to separate one band of frequencies from another. In other words, they perform well in the time domain, and poorly in the frequency domain. The frequency response can be improved slightly by cascading several stages. This can be accomplished in two ways. First, the signal can be passed through the filter several times. Second, the z-transform can be used to find the recursion coefficients that combine the cascade into a single stage. Both ways work and are commonly used. Figure (c) shows the frequency response of a cascade of four low-pass filters. Although the stopband attenuation is significantly improved, the roll-off is still terrible. If you need better performance in the frequency domain, look at the Chebyshev filters of the next chapter.